Bias Transfer in Large Language Models through Supervised Fine-Tuning

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Abstract

We examine how the bias of a training corpus transfers to a model fine-tuned on that corpus. Specifically, we fine-tune the Llama 3.2 3B Instruct model on four corpora that span a range of measured bias scores: NELA-GT (partisan US political news), NELA-GT-HB (a filtered subset of NELA-GT for high bias sources), NELA-PS (hyper-local auto-generated news content), and a corpus of Wikipedia articles on high schools in the US, which measured the least bias of the four. We evaluate the five resulting models (four fine-tuned conditions and the Llama base) on 300 completions across 15 prompts each covering a salient topic (making 1,500 total). Bias at the completions level is scored using an LLM-as-Judge rubric, and at the embedding-level with the Word Embedding Association Test (WEAT). Fine-tuning on the NELA-GT corpus raises mean completion bias above the base model, with the largest bias shifts specifically on the topics of climate and immigration; filtering the NELA-GT dataset to only include the high-bias sources raises bias further (approximately 1.5× the unfiltered corpus's shift over the base model), and also broadens the shift to topics that the unfiltered corpus left alone. Fine-tuning on the NELA-PS corpus reduces completion bias. However, fine-tuning on the Wikipedia corpus produces the highest mean completions bias, and the largest WEAT effect sizes, despite the corpus itself having the lowest bias scores. This Wikipedia result is attributable to three factors: overfitting on a small training set, hallucinations of fake content (laws, policy, etc.) that the judge attributes to high bias, and amplification of pre-existing biased associations in the base model. These results indicate that corpus-level bias scores are not reliable predictors of fine-tuned model bias, and that LLM-as-Judge evaluations that do not separate factual accuracy from biased framing will mark higher biases in models that produce more hallucinations.

1 Introduction

Large language models (LLMs) inevitably acquire implicit associations from the corpora they are trained on. Caliskan et al. [1] demonstrated that word embeddings derived from large text corpora reproduce human-like biases, for example associating flowers with pleasant attributes and insects with unpleasant ones. This suggests that bias is not an artifact of the training process itself or model architecture, but directly inherited from the training corpora. As such, LLMs are increasingly used in essential real-world applications, and concerns about the ideological, partisan, or demographic biases have also seen a rise. Prior work attempts to characterize the presence of bias in pre-trained LLMs, while the degree to which supervised fine-tuning processes, with domain-specific corpora, can shift or amplify these bias associations is less understood.

Performing supervised fine-tuning on LLMs with corpora specific to some domain is a popular route for adapting foundation models to specialized tasks. However, Wang et al. [2] demonstrate that fine-tuning an LLM on a corpus of US political news amplifies pre-existing right-leaning biases, with such biases intensifying across iterative synthetic training cycles. We seek to test this by designing a controlled experiment, with four unique corpora that span a bias gradient: 1. NELA-GT [3], a collection of partisan news articles with a mean LLM grade of 1.818 / 5; 2. NELA-PS [4], a corpus of hyper-local news dominated by auto-generated statistics, with a mean score of 0.212 / 5; and 3. a Wikipedia [5] derived corpus of high schools articles in the United States, scored at 0.060, serving as an essentially neutral baseline. We also filter out select articles from NELA-GT to obtain a fourth corpus, NELA-GT-HB (High Bias), keeping only the 76 sources with a representative per-source mean bias of $\geq 3/5$, which yields a corpus mean of 3.300 / 5. Each corpus is used to fine-tune an identical Llama 3.2 3B Instruct [6] base model, via Low-Rank Adaptation (LoRA) [7]. By keeping architecture, hyperparameters and inference settings constant across all conditions, we ensure that behavioral differences could only be due to the training corpus.

Bias is measured in two methods. Firstly, we apply an LLM-as-Judge pipeline to score completions generated by each of the five conditions on a 0–5 rubric, producing a distribution of bias scores across all five conditions. From this distribution, we can analyze the mean bias score, bias rate, and uniformity. Secondly, we apply the Word Embedding Association Test (WEAT) [1]. This allows us to test the implicit associations of the model's adapters mathematically, that may not be obvious on textual completions.

Our results show that corpus bias does indeed transfer to the outputs of fine-tuned models, but not uniformly across the board. Specifically, fine-tuning on partisan news shifts the overall bias distribution upwards, with the magnitude varying across topics. Filtered for high-bias articles, a model fine-tuned on such a corpus produces even more biased completions, and broadens this shift to topics that the model fine-tuned on the unfiltered corpus left unchanged. By contrast, the neutral Wikipedia condition produces a paradoxically large bias signal. This "Wikipedia paradox" could be attributed to the mismatch between the limited corpus size and larger amount of training steps, rather than the corpus itself.

2 Related Work

2.1 Bias in pre-trained models

Caliskan et al. [1] introduce the Word Embedding Association Test (WEAT), adapting the Implicit Association Test widely used in social psychology to quantify the association between target and attribute word sets to machine learning. Their analyses of GloVe embeddings trained on Common Crawl (an open source library of web crawl data) showed the same gender, racial, and age associations in WEAT that humans show in IAT studies. This shows that biases inherited from the corpora are measurable directly in the embedding space, and will be used as a mathematical tool to evaluate the embedding-level bias of a model.

2.2 Bias amplification during fine-tuning

Our work here is related to Wang et al. [2], who fine-tune GPT-2 models iteratively on a corpus of US political news and observe that political biases transfer not only from the corpus, but also compound and increase across successive training iterations. Their findings show that fine-tuning is not a neutral domain-adaptation step for LLMs, instead actively amplifying biases. We extend this question in several directions: 1. We use a more recent and modern instruction-tuned model (Llama 3.2 3B Instruct), rather than GPT-2; 2. We compare four corpora that each sit in a bias gradient (partisan news, its high-bias subset, hyper-local auto-generated content, and neutral encyclopedias), rather than a single partisan corpus; and finally, 3. We evaluate the resulting bias transfer with both an LLM-as-Judge rubric and WEAT, measuring surface and embedding-level associations.

Conversely, Srivastava et al. [8] pursue de-amplification of biases, showing that differential privacy during fine-tuning decreases bias transfer.

2.3 Biases in Wikipedia

Wikipedia is often considered the epitome of neutrality, and used as such for LLM training and evaluation. Navigli et al. [9] document that Wikipedia's editors are disproportionately male (87%) and English-speaking ($> 50\%$). This is directly relevant to our Condition N, which is fine-tuned on a corpus of Wikipedia articles on high schools in the United States, which scores as the most neutral ($\mu = 0.060$). Our results provide a clear example: the Wikipedia condition produces the largest WEAT effect sizes and some of the most biased completions, despite the apparent neutrality of its corpus. Superficial neutrality is not necessarily equal to being free from bias.

2.4 LLM-as-Judge Pipeline

Using an LLM to score the outputs of another model has become standard methodology for rubric-based evaluations at scale [10]. We adopt this method for scoring biases in the original corpus as a baseline, and also in LLM completions. In the process, we note an issue that the LLM-as-Judge does not fully resolve, which is when the judge encounters hallucinated or fabricated facts, such as a non-existent immigration policy, it cannot distinguish between hallucination and bias, and ends up labeling the hallucination as strong bias.

3 Methods

[TODO add pipeline figure]

3.1 Corpus Selection

We train and evaluate on four corpora.

3.1.1 NELA-GT

The NELA-GT corpus is sourced from the misinfo-general dataset [11], which takes a deduplicated aggregate of six NELA corpora [3] spanning from 2017 to 2022, compiling 4.2 million articles from 488 distinct publishers. Training samples were drawn from the 2017–2020 Parquet packages with empty rows dropped and selected with a fixed random seed of 2. 500,000 samples were taken and used as the training set for Condition GT.

3.1.2 NELA-GT-HB

The NELA-GT-HB (High Bias) corpus is constructed using NELA-GT as a baseline. We first scan the original GT corpus, and enumerate the number of articles represented by each source. A representative sample of 20 articles is selected from each source with $\geq 1{,}000$ articles, with grades from the original GT corpus audit reused where available, and additional articles graded to reach 20 per source. Filtering at the source level is necessary for the LLM judge: using API calls to grade all 4.2 million articles individually would require three orders of magnitude more Haiku calls. A mean bias score is taken for each source, and all articles from sources with ≥ 3 bias are chosen for the final corpus. This yields a total of 76 sources and 656,001 articles for NELA-GT-HB, of which 500,000 are randomly sampled for the training set via a deterministic hash. An independent validation of the final corpus confirms the high bias: as shown in Table 3, a 500 article sample scores a mean bias of 3.30, with 93.0% of articles having bias ≥ 2, compared to GT's 53.6%. We also use MBFC metadata as a cross-check only (it is publisher-level and US-centric, per the dataset README), which confirms the high-bias nature of the filtered set (as is shown in the source map Figure 1). Figure 9 shows an information fingerprint of all the sources considered; see Appendix G for the full source-selection figures, including the threshold choice (Figure 8).

3.1.3 NELA-PS

The NELA-PS corpus is a separate collection of 7.9 million articles from 1,093 sources, spanning March 2021 to January 2024 [4]. These sources are primarily dominated by Metric Media (6.99 million articles), a media network that produces articles with auto-generated local statistics that hold no editorial voice. This corpus is the training set for Condition PS. Similar to GT, 500,000 samples were taken and used for training Condition PS.

3.1.4 Wikipedia-derived

The Wikipedia corpus was built by recursively traversing a graph built on Wikipedia's category pages [5]. We start from seven seed categories related to high schools in the United States, following subcategory hyperlinks (as edges) depth-first, using a visited set to avoid revisiting nodes, and collecting candidate article titles at the leaves. Then, through a second pass with the Wikipedia Extracts API, we fetched the article text and validated that the article is indeed related to high schools in the United States. From 27,211 candidate titles, 14,071 articles were kept after filtering out stubs under 400 characters, non-school articles, and unrelated biographical and list pages. Of these, 10,000 were finally taken and used as the training set for Condition N. Sections irrelevant to bias, such as Alumni, References, See Also, etc. were stripped before saving. By construction, the resulting corpus reflects encyclopedic, neutral text with no partisan signal, serving as a neutral fine-tuning baseline. Figure 6 in the appendix reports per-token keyword frequencies across the school types represented in the corpus, indicating that the corpus is not topically uniform, despite aggregate neutrality.

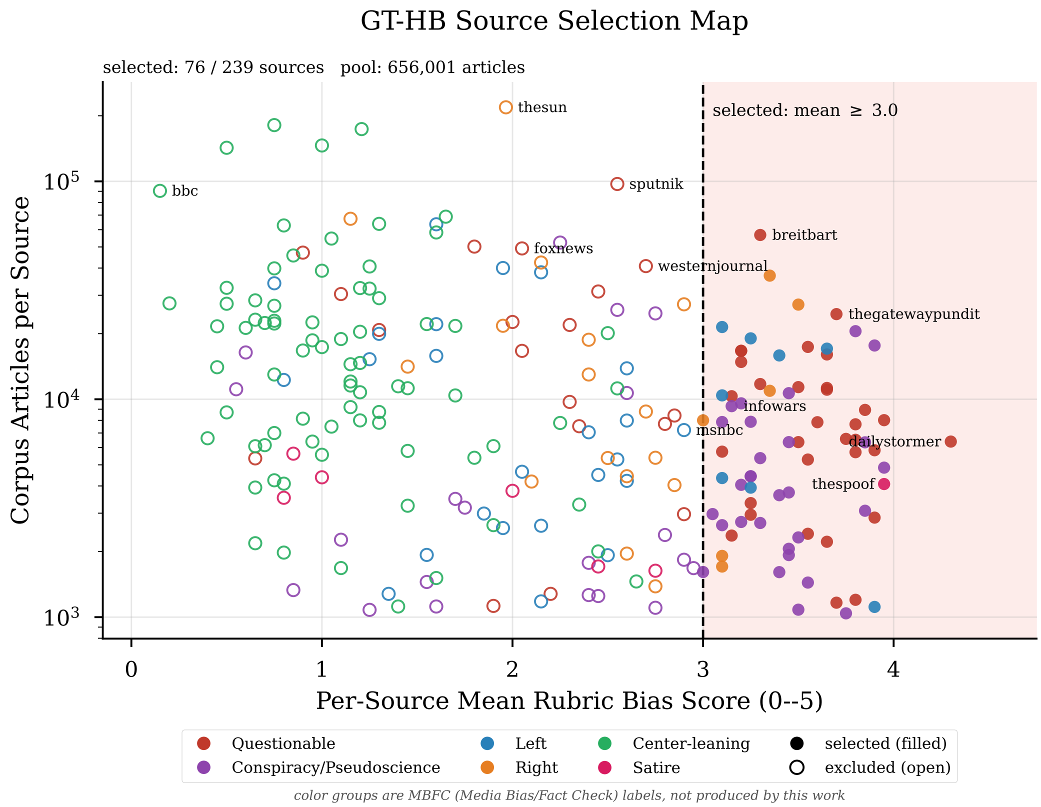

Figure 1: GT-HB source selection map. Each point is one audited NELA-GT source, plotting its per-source mean rubric score against its corpus article count (log scale). Filled markers are sources selected for the GT-HB whitelist (mean $\geq$ 3.0, vertical line); open markers are excluded; color gives the MBFC label group.

3.2 LLM-as-Judge Pipeline

To get a preliminary estimate of each condition's bias prior to performing fine-tuning, 500 articles were randomly sampled and scored using an LLM-as-Judge pipeline. This pipeline automates the bias scoring process, using Claude Haiku 4.5 (claude-haiku-4-5-20251001) [12] as a judge. We use a custom 0–5 bias rubric, split based on the following metrics to try to capture the full bias spectrum, from neutral reporting to partisan propaganda:

- 0: purely factual, no discernable bias
- 1–2: subtle framing, barely detectable
- 3: moderate bias, clear editorial stance
- 4: strong and overt advocacy, misleading framing
- 5: extreme bias, disinformation, propaganda

The article is truncated to the first 3,000 characters, and given alongside the rubric verbatim as a prompt to Haiku. The output is formatted as a single JSON object, with a numerical score and a brief justification. Parse failures are assigned a score of -1 and skipped.

Corpus bias scores are reported in Table 3. The four corpora span a broad bias gradient, making them suitable for the intended experimental contrast of a high-bias partisan vs. its higher-bias filtered subset vs. low-bias auto-generated vs. near-neutral dataset.

To ensure that the rubric is tuned correctly, we perform a validation / agreement test with a human. Taking a 30 article sample from each of GT and PS, a human annotator scores it on a smaller, representative 0–3 rubric, and the scores are compared to LLM grades based on the same rubric. Results are shown in Table 1.

Table 1: Human-LLM Agreement

Corpus	Exact Agreement	Within ± 1	Pearson r	LLM Drift
GT	67% (20/30)	97% (29/30)	0.803	+0.033
PS	53% (16/30)	100%	—	+0.333

Note: Pearson r not reported for NELA-PS, due to Haiku's scores being almost degenerate: 87% scoring 0 in the 30 article validation sample, therefore no meaningful conclusion. Validation conducted on the 0–3 rubric. Drift = mean(human) − mean(LLM)

To note is the Cohen's d [13] value for Haiku's scores (GT v. PS) is 1.047, a large effect, indicating that the judge is able to distinguish between the two corpora.

3.3 Training Setup

All fine-tuned conditions are derived from a single base model, Llama 3.2 3B Instruct [6], a 3 billion parameter instruction-tuned language model. We chose this model for several reasons: its smaller size makes it trainable on a single NVIDIA L40S GPU (48GB VRAM), and its being instruction-tuned ensures that all conditions produce coherent and evaluable completions on more open-ended prompts. This model with no fine-tuning serves as the base condition.

For fine-tuning, we use Low-Rank Adaptation (LoRA) [7]. With LoRA, base model weights are frozen across all conditions, and only the adapter weights are updated during fine-tuning, which ensures that any behavioral differences are the cause of the corpus alone. Adapters are applied to all seven projection layers, q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, and down_proj. We use $rank = 64$, $\alpha = 128$, and 5% dropout rate. All four fine-tuned conditions share the same model hyperparameters: learning rate of 3×10^{-4} with a cosine scheduler, batch size 4 with grad. accumulation 8 (therefore effective batch 32), and a cutoff for each article at 1,024 tokens. Total training steps and warmup steps vary across conditions to match token exposure across corpora of different sizes: 5,000 + 500 warmup for Conditions GT, GT-HB and PS, and 15,625 + 1,562 warmup for Condition N. Training for the NELA conditions was conducted with a RunPod L40S and Condition N was fine-tuned on a RunPod H100 SXM. Therefore, the conditions only differ in corpus and training size, as summarized in Table 2:

Table 2: Per-Condition training summary

Condition	Corpus	Samples	Steps	Final Loss
Base	—	—	—	—
GT	NELA-GT	500,000	5,000	1.855
PS	NELA-PS	500,000	5,000	0.572
N	Wikipedia	10,000	15,625	0.032
GT-HB	NELA-GT*	500,000	5,000	1.917

** NELA-GT corpus filtered for high bias articles; methodology described in Subsection 3.1.2.*

Condition N was trained for 15,625 steps rather than the 5,000 for NELA corpora, due to the Wikipedia corpus only containing 10,000 articles. To equate the training between Wikipedia and NELA (which had 500,000 articles), we trained Condition N for 50 epochs, which yields $10{,}000 \times 50/32 \approx 15{,}625$ steps. This step mismatch is a limitation: the longer training process exposes the adapters to more updates through the gradient, which, given the homogeneous training corpora, causes overfitting that may affect behavior independent of corpus.

The final training losses tell a similar story, showing different convergence patterns across the conditions (Figure 2). GT converged from 2.77 to 1.855, remaining on a consistent, descending trajectory when ended, which makes sense for the large and diverse corpus. GT-HB converged from 2.96 to 1.917, showing a descending trajectory similar to GT. PS converged from 3.67 to 0.572, approaching overfit by step 5,000. Wikipedia converged from 2.55 to 0.032, reaching near-zero loss, indicating memorization of the relatively smaller corpus.

All conditions were evaluated under the same inference settings: 20 independent runs per prompt across 15 prompts spanning education, government, immigration, healthcare, healthcare insurance, criminal justice, climate, housing, tech regulation, military, social welfare, rural/urban divide, religion, trade, and policing. Tokens generated were limited to 150, at a temperature of 0.7.

Compute. All fine-tuning used single-GPU cloud instances (RunPod). Conditions GT, GT-HB and PS each trained for 5,000 steps on one L40S (48 GB), taking 18.4 h, 8.51 h and 16.8 h of wall-clock time respectively; Condition N trained for 15,625 steps on one H100 SXM in 10.6 h, for a total of ≈54 GPU-hours across the four final runs. LLM-as-Judge scoring comprised roughly 9,080 Claude Haiku 4.5 API calls (1,500 corpus articles plus a 500-article

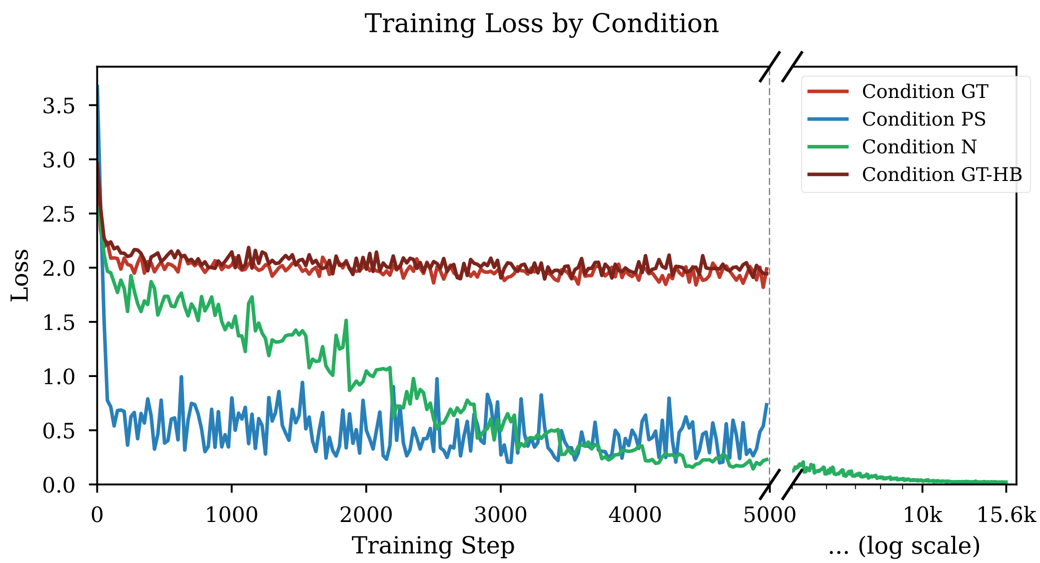

Figure 2: Training loss curves for Conditions GT, PS, N, and GT-HB. GT descends gradually over 5,000 steps; PS converges more steeply, approaching overfit by step 5,000; Wiki reaches near-zero loss by step 15,625; GT-HB descends from 2.96 to 1.92 on a GT-like trajectory.

GT-HB validation sample, roughly 4,320 articles graded for GT-HB source selection, 60 validation articles, 1,500 completions, and a 1,200-call second pass for hallucination flagging), each on inputs truncated to 3,000 characters. WEAT extraction and permutation tests run in under an hour on an M5 MacBook Air with 32GB ram. The full project consumed additional compute beyond these runs: shorter preliminary and aborted fine-tuning runs of each NELA condition, and early experiments on a smaller base model (Qwen2.5-1.5B), are not included in the totals above and are irrelevant for reproduction of results.

3.4 Evaluation Metrics

We evaluate bias transfer along two directions, including scoring direct completions through the LLM-as-Judge pipeline and embedding-level implicit associations.

3.4.1 Direct Completion Scoring

Each completion is scored by the same Claude Haiku 4.5 [12] judge pipeline as used for corpus validation, with an identical 0–5 rubric. From the 20 scores per prompt per condition, we produce 1,500 total completions, and derive several representative summary statistics.

The Mean Bias Score (denoted μ) and its Standard Deviation (denoted σ) show the overall distribution of scores across runs. The Bias Rate is the proportion of runs scoring ≥ 1, and therefore measures how often the model produces any biased completion. Uniformity is derived as Uniformity $= \frac{\mu}{\mu+\sigma}$. Ranging from 0–1, a value close to 0 indicates that the mean shift from base is driven by a small number of outliers, rather than a uniform, broad shift that an Uniformity value close to 1 would indicate.

3.4.2 WEAT

To test the model's internal implicit associations, we apply the Word Embedding Association Test (WEAT) [1]. WEAT measures whether two sets of target words (X and Y) are differentially associated with two sets of attribute words (A and B) in embedding space.

Unlike the original Caliskan et al. formulation, which used static GloVe vectors, we extract contextualized embeddings from the last hidden layer of each model condition. This is necessary because LoRA adapters modify the attention and MLP projection layers rather than the token embedding table, so static input embeddings would be identical across all conditions and produce no signal [7]. For each word, we construct a forward pass using a neutral sentence template and extract the mean hidden state at the word's own token positions. Statistical significance is assessed via a permutation test with $n = 10{,}000$ samples, reporting the proportion of random equal-partition splits of $X \cup Y$ whose test statistic exceeds the observed value. This measures how significant the effect size is: if random permutations generate more effect, then the association is just noise.

We evaluate on five tests designed to probe bias related to our corpora:

1. Test 1: High School Selectivity. Measures whether the model associates elite school terms (e.g. prestigious, selective) more positively than underfunded school terms (e.g. underfunded, neglected).

2. Test 2: Policy Necessity. Measures whether the model associates government policy terms (e.g. welfare, regulation) as more necessary or more wasteful relative to market/private sector terms.

3. Test 3: Immigration. Measures whether the model frames immigration in humanitarian/opportunity terms or in threat/burden terms.

4. Test 4: Economic Policy. Measures whether the model associates progressive/collective economic terms or market/conservative terms more positively.

5. Test 5: Media Trust. Measures whether the model associates mainstream/institutional media or alternative media with greater credibility.

For each test, effect size is computed as:

$$d = \frac{\bar{s}(X,\, A,\, B) - \bar{s}(Y,\, A,\, B)}{\mathrm{std}_{w \in X \cup Y}\, s(w,\, A,\, B)}$$

where

$$s(w,\, A,\, B) = \mathrm{mean}_{a \in A} \cos(w,\, a) - \mathrm{mean}_{b \in B} \cos(w,\, b)$$

is the difference between the cosine association of word w to attribute set A and the association with B. A positive d indicates that X is more associated with A than Y is.

4 Results

4.1 Corpus Bias

We score a 500 article random sample from each training corpus, using the same Haiku LLM-as-judge pipeline to characterize the training signal, prior to performing fine-tuning.

Table 3: Corpus bias scores across a 500 article random sample from each training corpus.

Corpus	Mean	$\%_0$	$\%_1$	$\%_2$	$\%_3$	$\%_4$	$\%_5$
GT	1.818	28.8	17.6	22.2	9.8	17.6	4.0
PS	0.212	94.4	0.6	1.2	0.2	0.4	3.2
Wiki	0.060	96.2	1.8	1.8	0.2	0.0	0.0
GT-HB	3.300	0.8	6.2	13.4	23.0	55.0	1.6

NELA-GT returned a mean score of 1.818 with a broad distribution across all bias scores; NELA-GT-HB has a mean of 3.30, with only 0.8% of articles scoring 0 and 55.0% scoring 4, essentially inverting the distribution shown in GT; NELA-PS returned a mean of 0.212 with 94.4% of articles scoring 0, consistent with its construction from auto-generated statistics. Finally, Wikipedia scored an even lower mean of 0.060, also consistent with the platform's focus on neutrality. The full bias score distributions for the four corpora are shown in Figure 7.

4.2 Completions Bias

Completions bias scores across all conditions are shown in Table 4 and visualized in Figure 3.

Table 4: Completions bias scores across all 15 prompt topics ($n = 300$ per condition).

Condition	Mean	$\%_0$	$\%_1$	$\%_2$	$\%_3$	$\%_4$	$\%_5$
Base	0.847	23.3	68.7	8.0	0.0	0.0	0.0
GT	1.470	24.0	19.3	46.7	7.0	1.7	1.3
PS	0.507	69.0	16.3	11.3	2.0	1.0	0.3
N	2.153	13.3	20.7	35.3	8.3	12.7	9.7
GT-HB	1.763	20.0	15.3	43.3	12.0	8.3	1.0

Condition PS produces the lowest mean bias score (0.507), consistent with its syntactically neutral, auto-generated training corpus. Condition N produces the highest mean (2.153), driven by elevated scores on immigration, religion, and trade. Condition GT shows a substantial bias increase compared to Base (1.470 v. 0.847, a 73% increase). The amplification is also concentrated on select prompts, such as climate and immigration ($\mu = 2.350$ and $\mu = 2.050$, respectively). Condition GT-HB has a mean of 1.763, around 1.5 times GT's lift over the base. By topic, GT-HB shows the largest increases on education (+1.00) and tech regulation (+1.00), and near-zero gain on GT's strongest biased topics (climate +0.05 and military +0.10). Per-topic results are shown in Table 6.

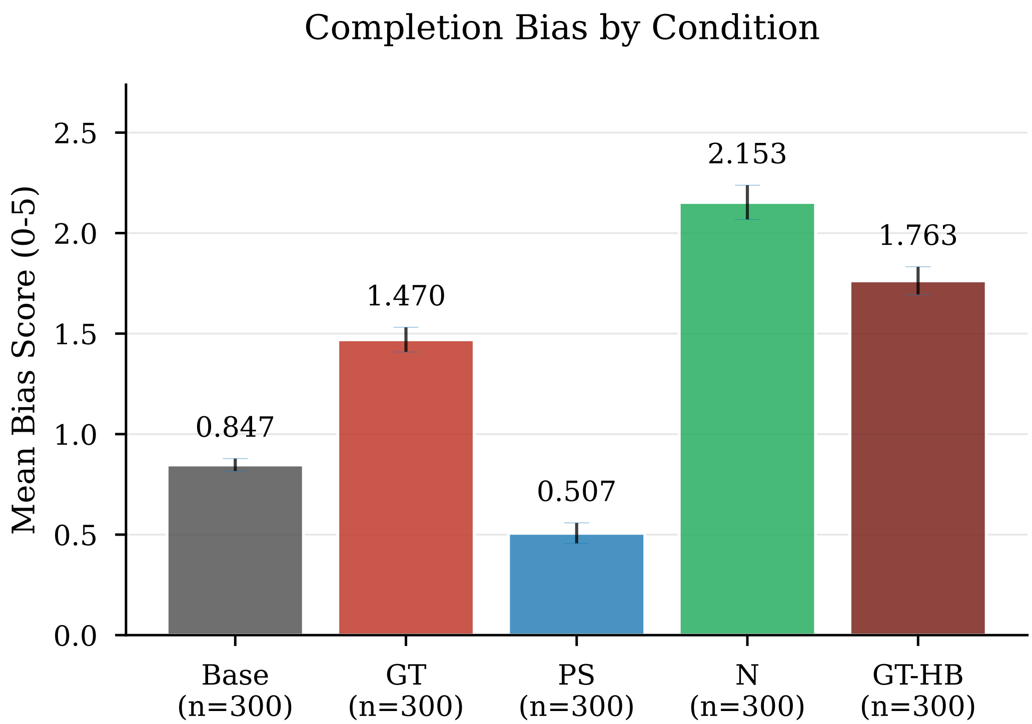

Figure 3: Mean completion bias score by condition across all 1,500 completions (error bars: ± 1 SE, $n = 300$ per condition).

4.3 WEAT Results

Table 5: WEAT Results

Metric	Base	GT	PS	N	GT-HB
Test 1 d	1.150	0.866	0.825	1.435	0.540
... p	0.012	0.045	0.052	0.001	0.149
Test 2 d	-1.314	-1.408	-1.402	-0.973	-1.376
... p	0.997	0.997	0.997	0.970	0.997
Test 3 d	0.911	0.890	0.955	1.632	1.190
... p	0.008	0.021	0.005	0.0001	0.001
Test 4 d	-0.555	-0.430	-0.534	-0.491	-0.449
... p	0.859	0.812	0.856	0.841	0.823
Test 5 d	0.575	0.727	0.911	1.088	0.579
... p	0.139	0.081	0.037	0.014	0.136

WEAT results are shown in Table 5 and Figure 4. Test 3 produces significant results for all conditions, and Test 1 for Conditions Base, GT, and N, while Test 2 and 4 are null. Test 5 yields significant results, but only for Conditions PS and N.

Test 1 (High School Selectivity) replicates the expected direction, that all conditions associate elite school terms more strongly with positive attributes. However, compared to Base, fine-tuning on NELA-GT and NELA-PS slightly weakens this association, as ($\Delta_{\text{GT}} = -0.284$ and $\Delta_{\text{PS}} = -0.325$). Condition GT-HB weakens it furthest ($\Delta_{\text{GT-HB}} = -0.610$), to the point of non-significance ($d = 0.540$ with $p = 0.149$). Condition N, however, strengthens this association ($\Delta_{\text{N}} = +0.285$ with $p = 0.001$), showing an early instance of the Wikipedia paradox at the embedding level, and extending a consistent contrast: news fine-tuning weakens the selectivity association, while Wikipedia fine-tuning strengthens it.

Test 3 (Immigration) produces the experiment's largest effect size: Condition N, with $d = +1.632$ with $p < 0.001$. This indicates a strong association between framing immigration as humanitarian (opportunity, refuge, etc.) and positive attributes. This also converges with the completion score results, where Condition N also produces the highest immigration bias rate (of 100%) and mean score (of 4.05). Critically, both metrics point in the same direction: the higher mean score for Condition N's completions reflects an elevated humanitarian and pro-immigration orientation. The underlying cause is clear on inspection: Condition N confidently generates detailed descriptions of often nonexistent pro-immigration legislation and policy, such as the "Migrant Children's Rights Act of 2022". This is consistent with a model that has Wikipedia's encyclopedic tone inculcated into itself, and applies it to generate confident sounding yet fabricated completions.

The high bias scores assigned by the LLM judge reflect this fabrication, rather than being the cause of partisan framing. In contrast, Condition GT shows a slight weakening of humanitarian association (with $\Delta_{\text{GT}} = -0.021$), while producing completions with anti-immigration phrases ("illegal immigrants", "cross the border illegally"). Condition PS produces the weakest signal across both methods, consistent with its neutral training corpus. Condition GT-HB, by contrast, amplifies the humanitarian association beyond Base,

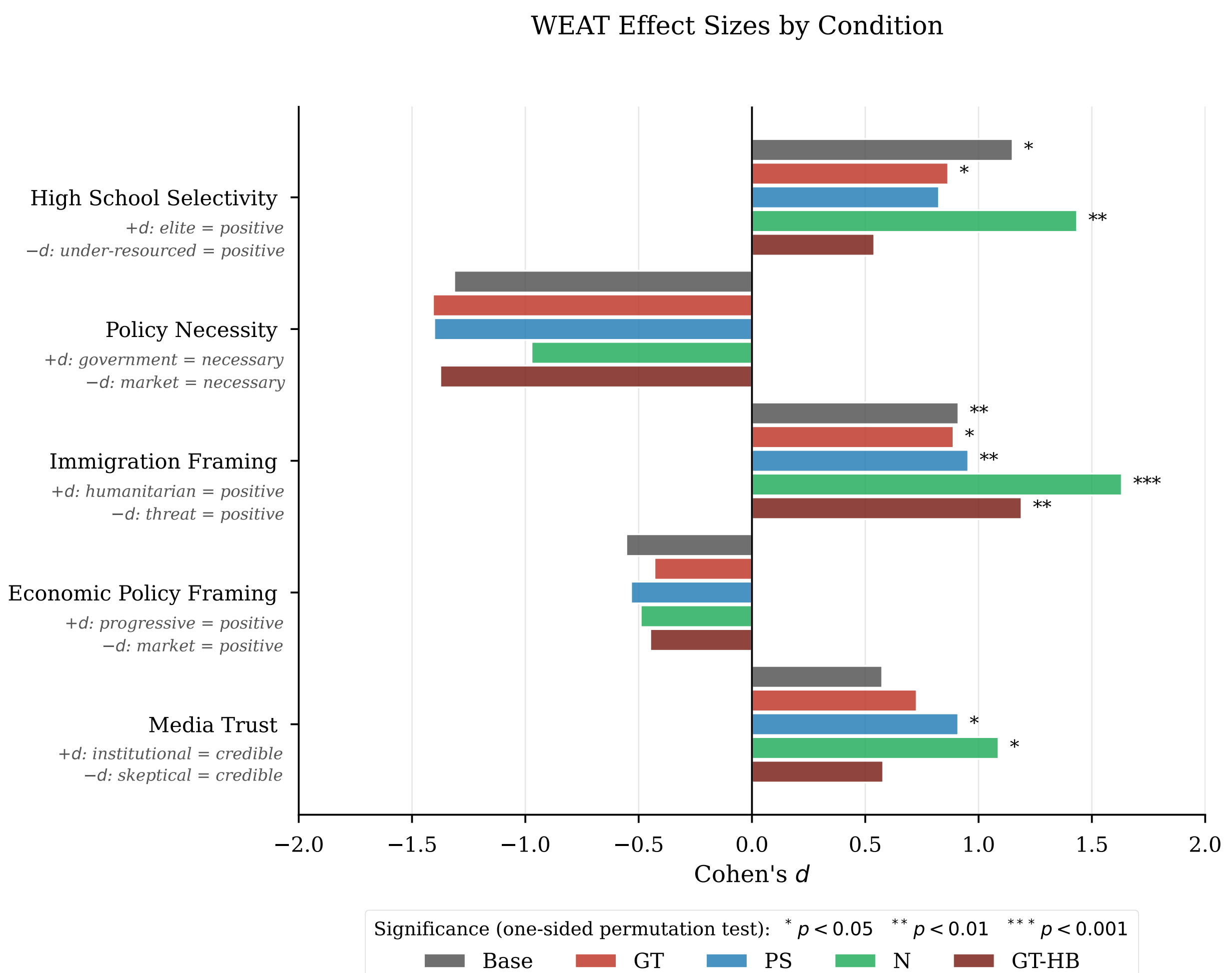

Figure 4: WEAT effect sizes (Cohen's d) by condition and test, with significance markers from the one-sided permutation test.

GT, and PS ($d = +1.190$ with $p = 0.001$), second only to Condition N. Unlike Condition N's hallucination-driven effect, this amplification coexists with a hallucination rate of only 12.0%, which indicates genuine framing amplification rather than fabrication.

Test 5 (Media Trust) yields significant effects for Condition PS ($d = +0.911$ with $p = 0.037$) and Condition N ($d = +1.088$ with $p = 0.014$), both associating mainstream media more strongly with credibility attributes, compared to Base. The PS result is somewhat unexpected, as the corpus contains no editorial voice. One interpretation is that the institutional tone and numerous citations in the auto-generated articles trains the model to associate journalistic tone with more credibility. Somewhat surprisingly, Condition GT-HB shows no shift in media-trust association despite its alternative-media-heavy corpus: its effect ($d = 0.579$ with $p = 0.136$) is nearly identical to Base ($d = 0.575$).

4.4 Variance Analysis

A per-prompt heatmap of mean bias rate is reported in the Appendix at Figure 5. Table 6 summarizes five representative topics, with the full table at Table 8.

5 Discussion

5.1 GT and PS Bias Transfer

Condition GT produces a mean completion bias of $\mu = 1.470$, significantly higher than the Base condition, which has $\mu = 0.847$. However, the bias increase is not general: GT actually has elevated biases on specific topics, such as climate ($\mu = 2.350$) and immigration ($\mu = 2.050$), while others (such as housing and policing) show minimal increases. Therefore, fine-tuning on a partisan corpus shifts the model's bias distribution upwards, with the magnitude higher and concentrated on topics with representation in the NELA-GT corpus.

Condition PS produces the lowest mean of all conditions ($\mu = 0.507$), with 69.0% of completions scoring perfect 0. This is consistent with the near-zero bias of the corpus itself.

Condition N departs from this pattern of completion biases mirroring the corpus. Despite its training corpus scoring the lowest of all four ($\mu = 0.060$), it produces the highest mean completion bias ($\mu = 2.153$). Immigration, religion, social welfare, and trade all exceed Uniformity > 0.5 (see Table 8) while also carrying the largest mean shifts of any condition on those topics, a combination unique to Condition N. We examine this contradiction below.

5.2 Filtering the Corpus for Bias (GT-HB)

Condition GT-HB tests whether concentrating the partisan corpus's bias intensifies its bias transfer. We see the result as a broadening of bias rather than an amplification: GT-HB's mean completion bias exceeds GT's on 13 of 15 topics (see Table 8), yet the smallest gains are on the topics GT already elevated most (climate +0.05, military +0.10). This suggests that there exists a ceiling, where the largest gains appear on topics the unfiltered corpus left largely unaffected (education +1.00, tech regulation +1.00).

Table 6: Per-prompt variance analysis for five representative topics (full results in Table 8).

Prompt	Condition	Mean	Std	Bias Rate	Uniformity
Immigration	Base	1.100	0.553	90%	0.666
...	GT	2.050	1.050	95%	0.661
...	PS	1.250	1.164	65%	0.518
...	N	4.050	1.050	100%	0.794
...	GT-HB	2.550	0.686	100%	0.788
Climate	Base	0.900	0.447	85%	0.668
...	GT	2.350	0.933	100%	0.716
...	PS	0.900	1.334	40%	0.403
...	N	4.350	0.875	100%	0.833
...	GT-HB	2.400	1.231	95%	0.661
Crim. Justice	Base	1.050	0.605	85%	0.635
...	GT	1.350	0.988	70%	0.577
...	PS	0.550	0.826	35%	0.400
...	N	1.950	0.999	90%	0.661
...	GT-HB	1.050	1.276	45%	0.451
Religion	Base	0.850	0.366	85%	0.699
...	GT	1.600	1.314	80%	0.549
...	PS	0.300	0.733	20%	0.291
...	N	2.050	0.999	100%	0.672
...	GT-HB	1.950	1.276	85%	0.604
Trade	Base	1.000	0.324	95%	0.755
...	GT	1.450	1.191	75%	0.549
...	PS	0.300	0.657	20%	0.313
...	N	3.050	1.050	100%	0.744
...	GT-HB	1.300	0.865	75%	0.601

We see two reasons behind this broadening. The first is a shift in the composition of the corpus: the 76 sources selected that survive filtering are predominantly hyper-partisan outlets that politicize every topic they cover, rather than outlets with narrow topical focuses. The second is dilution removal: 93.0% of GT-HB training articles score ≥ 2 on the bias rubric (taken from the 500 article validation), against 53.6% for GT (see Table 3), so the low-bias articles that may dilute the partisan signal in GT are largely absent.

Unlike Condition N's lift, GT-HB's survives hallucination cleaning: its hallucination-cleaned mean of 1.549 still exceeds GT's clean mean of 1.291 (see Table 7). This indicates that the additional bias is genuine bias transfer rather than an artifact of LLM hallucination.

One problem remains, which is that the filter concentrates the corpus into 76 sources, and even further with how the ten largest sources account for 39.4% of the sampling pool, so part of the broadened bias may reflect the transfer to a narrower stylistic voice of each individual source. We talk more about this problem in the Limitations section.

5.3 The Wiki Paradox

Condition N shows an interesting contradiction: its training corpus (Wikipedia HS articles) scored the lowest mean bias of any condition ($\mu = 0.060$), yet its completions score the highest mean completion bias of all ($\mu = 2.153$) and the strongest non-null WEAT effect sizes. We propose three possible explanations for this.

Overfitting with excess training steps. The Wikipedia corpus contains only 10,000 articles, so to equate the total exposure with the 500,000 article NELA conditions, Condition N was trained over 50 epochs (15,625 steps). The near-zero final loss of 0.032 confirms near complete memorization of the training data. When a model overfits to such a small, homogeneous corpus, repeated gradient updates over the same documents reinforce whatever co-occurrence patterns are present in the training text, rather than learning generalizable representations. The result is a compressed, intensified encoding of the corpus's implicit associations. This is consistent with Wang et al. [2], who show that iterative fine-tuning amplifies pre-existing political associations even when the training data appears neutral. Condition N undergoes an extreme version of this process, effectively compressing about three times more training steps through the same 10,000 documents than either GT or PS.

Hallucination-driven score inflation. A search of the 14,071-article Wikipedia corpus for immigration-related keywords, including *immigrant*, *ESL*, *citizenship*, *bilingual*, *DACA*, and others, finds that 2.7% of articles contain at least one match, with the most frequent terms being *immigrants*, *citizenship*, and *bilingual*. A full list of keywords and match counts is in the appendix, Table 9.

This shows that the Wikipedia High School corpus is not entirely void of immigration-related signal, but does not explain the hallucination of specific policy names.

To quantify the contribution of fabricated content directly, we re-scored every completion in the 1,500 completion set (GT-HB was built after, so only one pass was done) with an additional binary flag for hallucination, which is set whenever the completion references a fact (law, policy act, and others) that does not exist. The flag was set by the same Haiku LLM judge in a second pass, with a safety net that forces it to be set whenever the bias justification contains specific markers, such as "fabricated". Table 7 reports the hallucination rate and the change in bias score attributable to hallucination. Notably, Condition N hallucinates on

44.3% of completions, and on all completions from Climate and Immigration prompts (see Figure 5, panel B for full hallucination rates).

With hallucinated completions dropped, Condition N's mean bias drops from 2.153 to 1.246, a reduction of 0.908 (42%). The other Conditions move by at most 0.214. The mean bias for the clean set of Condition N completions is 1.246, in fact below the clean average for GT-HB (which is 1.549, highest of all). Once hallucinated content (which is an artifact intrinsic to all LLMs, not specifically to the corpus or fine-tuning step) is removed, the condition tuned on partisan corpus instead produces more biased completions than the Wikipedia-tuned condition, totally reversing the pattern observed in Table 4.

The specific breakdown of the hallucinations per Condition, per topic can be found in Table 10 in the Appendix. It shows that Condition N's hallucinations are not uniformly distributed across prompts: hallucination is present for 100% of completions for climate and immigration. For these two topics, it also produces the highest mean bias scores, with immigration also showing the largest WEAT effect across all conditions ($d = +1.632$). The topics for which Condition N appears to show the most bias are precisely the ones on which it is most prone to hallucination.

Taken together, these three observations indicate that Condition N's apparent bias is the joint output of three mechanisms acting in sequence rather than a single transfer of corpus content.

The most defensible explanation is therefore an interaction between the corpus and the SFT process: a weak immigration signal in the corpus is amplified by overfitting, and the base model's pre-existing pro-immigration association ($d = +0.911$ at baseline; see Table 5) supplies the fake legislative content.

This is further compounded by an artifact of the LLM-as-Judge pipeline. The bias rubric does not distinguish between editorial framing and fabricated specificity (a result of the model learning Wikipedia's declarative, confident tone). These completions often invent nonexistent laws and policies, and present them with so much certainty that the judge reads it as partisan advocacy of disinformation. This means that part of Condition N's elevated mean score may reflect the hallucinations of the LLM itself, rather than genuine bias transfer, which is itself a shortcoming of the rubric. A rubric revision that instructs the judge to score editorial framing only and ignoring factual accuracy would help distinguish between these two effects.

Table 7: Hallucination rate and bias score inflation per condition. μ_{all} is the Table 4 value across all 300 completions; μ_{clean} excludes hallucination-flagged completions; $\Delta\mu_{hall.}$ is the bias difference attributable to hallucination.

Condition	Hall. Rate	μ_{all}	μ_{clean}	$\Delta\mu_{hall.}$
Base	2.0%	0.847	0.830	+0.017
GT	10.7%	1.470	1.291	+0.179
PS	5.3%	0.507	0.384	+0.123
N	44.3%	2.153	1.246	+0.908
GT-HB	12.0%	1.763	1.549	+0.214

Amplification of pre-existing bias. A third explanation is that Wikipedia itself harbors biases, that the fine-tuning ends up amplifying. Navigli et al. [9] document several structural biases in Wikipedia: its article distribution is significantly skewed toward the Sports, Music, Places, and Politics domains; its editors are disproportionately male (87%) and English-speaking (over 50%); this demographic inevitably shapes both the content and implicit biases in the articles. Wang et al. [2] provide evidence of this amplification: fine-tuning on balanced or center-leaning corpora intensifies pre-existing associations, with bias increasing across successive fine-tuning iterations.

Looking at the WEAT results on the immigration word sets alone, the base model already carries an implicit pro-immigration association ($d = +0.911$), and fine-tuning on Wikipedia amplifies it to $d = +1.632$, the largest single effect size across all conditions and word sets.

These explanations need not be mutually exclusive. Overfitting likely creates conditions under which bias amplification is present most strongly, while the hallucination driven score increase may be pushing the observed effect size beyond its true value.

6 Limitations

6.1 Condition N Training Step Too High

The four fine-tuned conditions are not equal in their training times. Conditions GT, GT-HB and PS were trained for 5,000 steps on 500,000 articles each, while Condition N was trained for 15,625 steps on 10,000 articles each, to match the total token exposure. This means that Condition N's adapters received roughly three times as many gradient updates, a source that would shift the model's behavior that is not present in the other three conditions. Discussed effects on Condition N in Section 5.3.

6.2 Overfitting on Conditions PS and N

The final training losses indicate that Conditions PS (0.572) and N (0.032) are approaching memorization of the training data. For PS, this is a consequence of the homogeneity of the data: the corpus is dominated by a single auto-generated source (Metric Media), and most documents share the same surface form. For N, the small corpus size forces training with more epochs to match the token exposure. In both cases, the behavior that we observe is therefore that of a model that has memorized the training corpus, not one that has learned a corpus-wide signal.

6.3 GT-HB Source Concentration

The GT to GT-HB filter narrows 488 sources to only 76, with the top 10 sources taking up 39.4% of the pool. Therefore, the broadened bias compared to GT cannot be generally attributed to the higher level of bias; some of the change may be due to a narrowing of editorial tone caused by a narrowing of the sources. Whether the change is due to a mere stylistic change can be tested by taking a random sample of 76 GT sources and analyzing a control model trained from it; this is left for future work.

6.4 LLM-as-Judge Unable to Distinguish Hallucination from Bias

The LLM-as-Judge rubric we use scores completions on the biased tone and content together, so the judge cannot separate the model being truly biased and the model just saying something false. For most conditions this issue is somewhat trivial, but for Condition N, 44.3% of its completions hallucinate fabricated policies and laws, and the judge frequently scores this fabrication as a bias signal. We show the impact directly via cleaning out hallucinated completions and re-calculating a mean score based on the clean entries (see Table 7), and find that this clean score reverses the bias rank between GT and N.

6.5 Single Model

All conditions fine-tune from the same Llama 3B 3.2 Instruct base model. This was chosen for its smaller size and therefore lower training and inference compute costs, but it is at the very small end of instruction-tuned models and especially compared to other foundation models. Different base models, especially those with different architectures (Qwen, Mistral, Gemma, etc.) may respond to the corpus differently. Cross-model replication is the natural next step.

7 Conclusion

We set out to test whether the bias amplification observed by Wang et al. [2] for partisan corpora extends similarly to a broad bias gradient. Specifically, we investigate whether corpora of varying measured bias produce fine-tuned models of correspondingly varied bias. We fine-tuned a 3B instruction-tuned LLM on four corpora each located on a bias gradient and evaluated the resulting completions across domains with an LLM-as-Judge rubric and the Word Embedding Association Test (WEAT) [1]. The hypothesis is partially supported, as fine-tuning on partisan news (GT) transferred bias to completions on specific topics, with climate and immigration receiving the largest shifts, while fine-tuning on the almost neutral PS corpus reduced bias, as expected. For the filtered GT-HB corpus, the filtering process to its most biased sources strengthens and broadens the bias transfer to more topics, rather than increasing bias further on hot spots; this suggests the existence of a ”ceiling”. This effect survives cleaning from hallucination. The hypothesis breaks on Condition N, derived from Wikipedia articles. Despite producing the lowest corpus bias scores, it generates completions with the highest mean bias and the largest WEAT effect sizes. This is a paradox attributable to overfitting on a small homogeneous corpus, hallucinations that the judge sees as bias, and amplification of pre-existing bias in the base model [2]. In general, corpora that appear neutral on the surface can still produce strongly biased fine-tuned models when training develops into memorization, and bias evaluation pipelines that do not separate factual accuracy from biased framing cannot distinguish between hallucinations and actual bias, resulting in overcounting biases that are actually just classic LLM hallucination.

A Per-Prompt Heatmaps

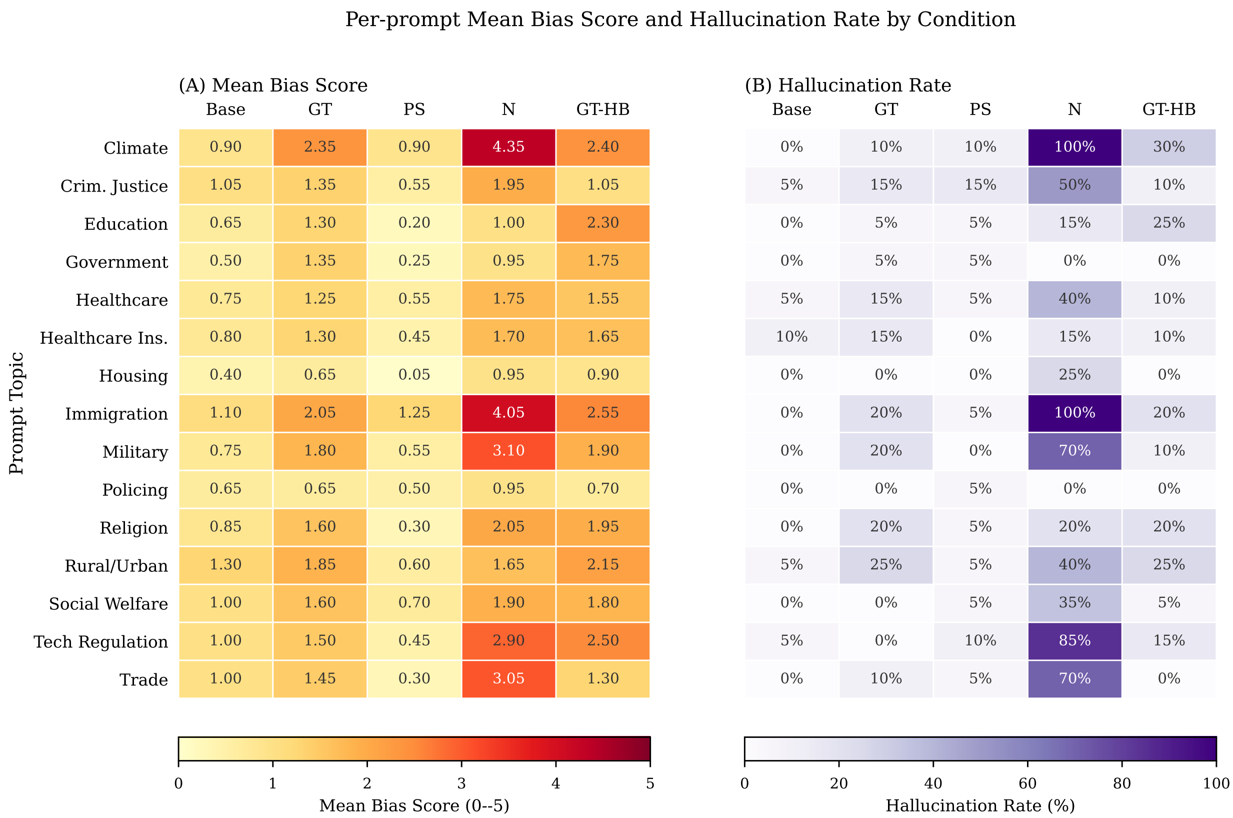

Figure 5: Per-prompt heatmaps: (A) is Mean bias score, (B) is Hallucination rate, all run per condition and prompt topic.

B Full Variance Analysis

Table 8: Per-prompt variance analysis across all 15 topics and 5 conditions ($n = 20$ runs each).

Prompt	Condition	Mean	Std	Bias Rate	Uniformity
Climate	Base	0.900	0.447	85%	0.668
...	GT	2.350	0.933	100%	0.716
...	PS	0.900	1.334	40%	0.403
...	N	4.350	0.875	100%	0.833
...	GT-HB	2.400	1.231	95%	0.661
Crim. Justice	Base	1.050	0.605	85%	0.635

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Table 8 continued

Prompt	Condition	Mean	Std	Bias Rate	Uniformity
...	GT	1.350	0.988	70%	0.577
...	PS	0.550	0.826	35%	0.400
...	N	1.950	0.999	90%	0.661
...	GT-HB	1.050	1.276	45%	0.451
Education	Base	0.650	0.489	65%	0.570
...	GT	1.300	1.174	65%	0.525
...	PS	0.200	0.696	10%	0.223
...	N	1.000	0.858	70%	0.538
...	GT-HB	2.300	1.218	90%	0.654
Government	Base	0.500	0.513	50%	0.494
...	GT	1.350	1.226	65%	0.524
...	PS	0.250	1.118	5%	0.183
...	N	0.950	0.759	70%	0.556
...	GT-HB	1.750	1.410	75%	0.554
Healthcare	Base	0.750	0.444	75%	0.628
...	GT	1.250	0.910	75%	0.579
...	PS	0.550	0.686	45%	0.445
...	N	1.750	0.716	95%	0.710
...	GT-HB	1.550	1.050	80%	0.596
Healthcare Ins.	Base	0.800	0.616	70%	0.565
...	GT	1.300	1.031	70%	0.558
...	PS	0.450	0.686	35%	0.396
...	N	1.700	0.923	90%	0.648
...	GT-HB	1.650	1.040	80%	0.613
Housing	Base	0.400	0.503	40%	0.443
...	GT	0.650	0.875	40%	0.426
...	PS	0.050	0.224	5%	0.183
...	N	0.950	1.099	55%	0.464
...	GT-HB	0.900	1.119	45%	0.446
Immigration	Base	1.100	0.553	90%	0.666
...	GT	2.050	1.050	95%	0.661
...	PS	1.250	1.164	65%	0.518
...	N	4.050	1.050	100%	0.794
...	GT-HB	2.550	0.686	100%	0.788
Military	Base	0.750	0.550	70%	0.577
...	GT	1.800	0.616	95%	0.745
...	PS	0.550	0.826	35%	0.400
...	N	3.100	1.651	95%	0.652

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Table 8 continued

Prompt	Condition	Mean	Std	Bias Rate	Uniformity
...	GT-HB	1.900	0.912	95%	0.676
Policing	Base	0.650	0.489	65%	0.570
...	GT	0.650	0.875	40%	0.426
...	PS	0.500	0.688	40%	0.421
...	N	0.950	0.826	65%	0.535
...	GT-HB	0.700	0.733	55%	0.489
Religion	Base	0.850	0.366	85%	0.699
...	GT	1.600	1.314	80%	0.549
...	PS	0.300	0.733	20%	0.291
...	N	2.050	0.999	100%	0.672
...	GT-HB	1.950	1.276	85%	0.604
Rural/Urban	Base	1.300	0.571	95%	0.695
...	GT	1.850	1.040	90%	0.640
...	PS	0.600	1.095	25%	0.354
...	N	1.650	1.182	80%	0.583
...	GT-HB	2.150	0.988	95%	0.685
Social Welfare	Base	1.000	0.562	85%	0.640
...	GT	1.600	0.754	85%	0.680
...	PS	0.700	0.733	55%	0.489
...	N	1.900	0.912	90%	0.676
...	GT-HB	1.800	1.005	85%	0.642
Tech Regulation	Base	1.000	0.324	95%	0.755
...	GT	1.500	0.607	95%	0.712
...	PS	0.450	0.826	30%	0.353
...	N	2.900	1.373	100%	0.679
...	GT-HB	2.500	1.235	100%	0.669
Trade	Base	1.000	0.324	95%	0.755
...	GT	1.450	1.191	75%	0.549
...	PS	0.300	0.657	20%	0.313
...	N	3.050	1.050	100%	0.744
...	GT-HB	1.300	0.865	75%	0.601

C Wikipedia Corpus Immigration Keyword Counts

Table 9: Immigration-related keyword matches in the 14,071-article Wikipedia high-school corpus. "Articles" is the number of distinct articles containing ≥ 1 occurrence; "Occurrences" is the total raw count across the corpus.

Keyword	Articles	% of corpus	Occurrences
immigrants	86	0.61%	111
citizenship	69	0.49%	74
bilingual	57	0.41%	79
immigrant	52	0.37%	64
ESL	50	0.36%	63
immigration	29	0.21%	36
refugee	17	0.12%	27
ICE	16	0.11%	35
sanctuary	15	0.11%	19
English language learner	14	0.10%	14
asylum	13	0.09%	16
migrant	10	0.07%	12
deportation	5	0.04%	6
foreign-born	5	0.04%	6
undocumented	4	0.03%	6
visa†	4	0.03%	4
border	4	0.03%	10
DACA	2	0.01%	2
Title III	2	0.01%	2
naturalization	2	0.01%	2
migrants	1	0.01%	1
DREAMer	1	0.01%	1
Any keyword	**382**	**2.7%**	—

† Higher false-positive risk; interpret counts carefully.

D Full Hallucination Summary per Prompt

Table 10: Hallucination rate per prompt topic and condition (% of 20 runs flagged as hallucinated).

Prompt	Base	GT	PS	N	GT-HB
Climate	0.0%	10.0%	10.0%	**100.0%**	30.0%
Criminal justice	5.0%	15.0%	15.0%	50.0%	10.0%
Education	0.0%	5.0%	5.0%	15.0%	25.0%

Prompt	Base	GT	PS	N	GT-HB
Government	0.0%	5.0%	5.0%	0.0%	0.0%
Healthcare	5.0%	15.0%	5.0%	40.0%	10.0%
Healthcare insurance	10.0%	15.0%	0.0%	15.0%	10.0%
Housing	0.0%	0.0%	0.0%	25.0%	0.0%
Immigration	0.0%	20.0%	5.0%	**100.0%**	20.0%
Military	0.0%	20.0%	0.0%	70.0%	10.0%
Policing	0.0%	0.0%	5.0%	0.0%	0.0%
Religion	0.0%	20.0%	5.0%	20.0%	20.0%
Rural urban	5.0%	25.0%	5.0%	40.0%	25.0%
Social welfare	0.0%	0.0%	5.0%	35.0%	5.0%
Tech regulation	5.0%	0.0%	10.0%	85.0%	15.0%
Trade	0.0%	10.0%	5.0%	70.0%	0.0%

E Wikipedia Keyword Frequencies by School Type

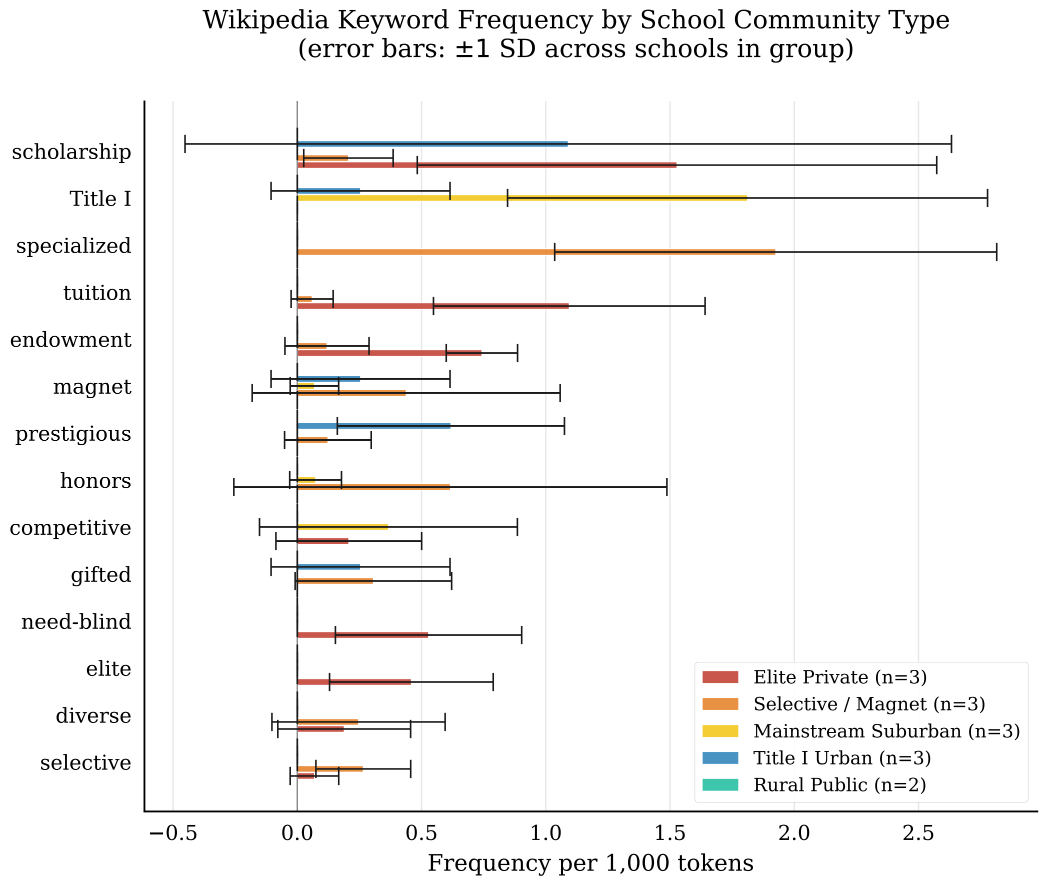

Figure 6: Wikipedia keyword frequency by school type (per 1,000 tokens, error bars ±1 stddev across schools within each group). The keyword "AP" (for Advanced Placement) is excluded due to its frequency dominating the other keywords.

F Training Corpus Bias Score Distributions

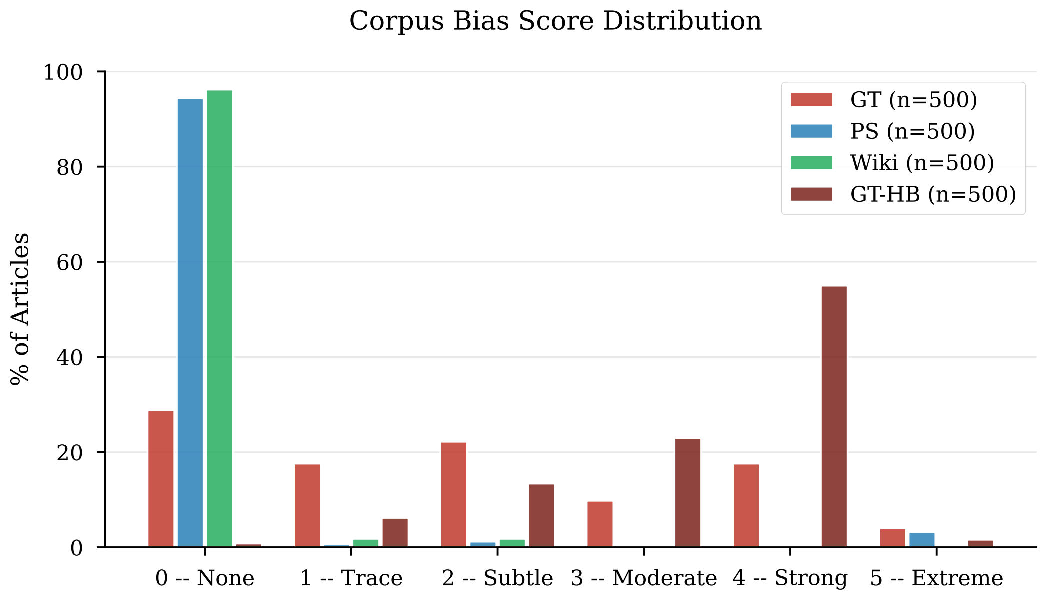

Figure 7: Distribution of bias scores (0–5) across the four training corpora.

G GT-HB Source Selection

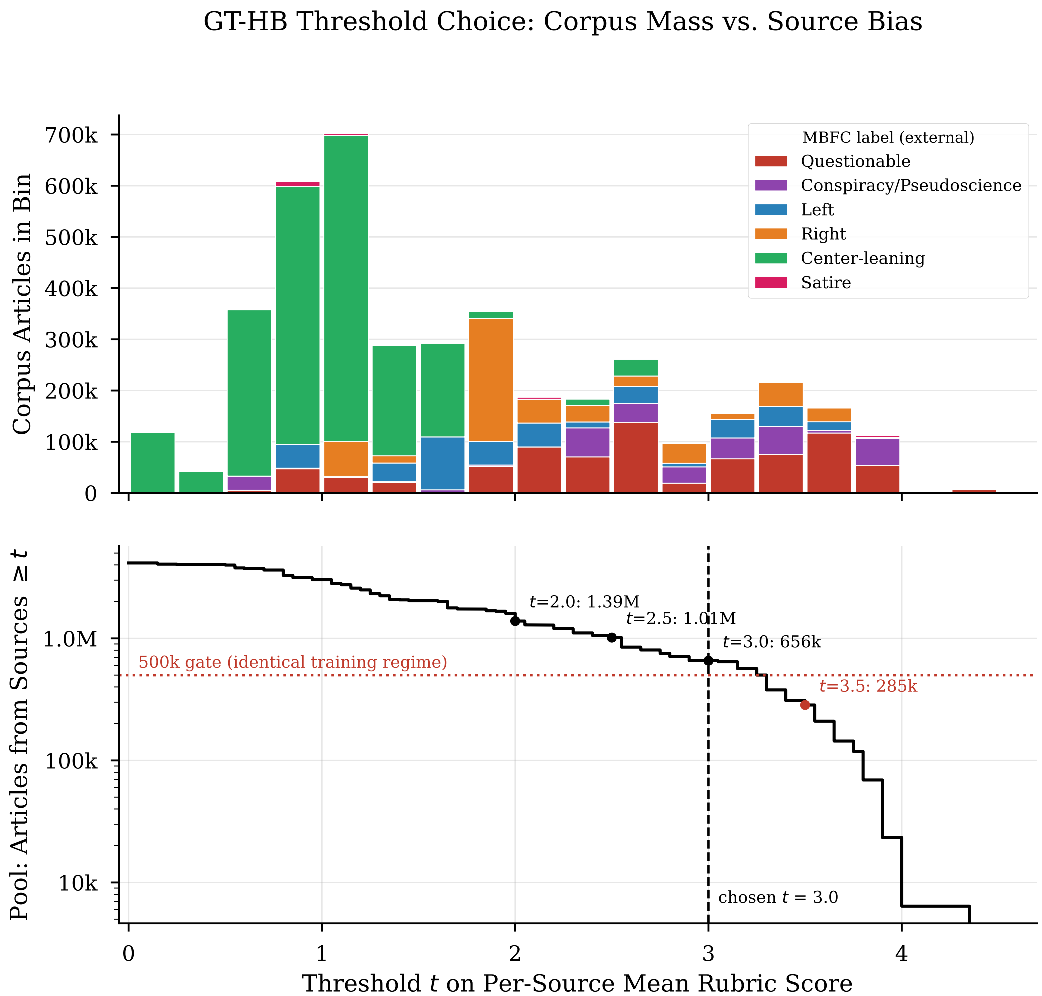

Figure 8: GT-HB threshold choice. Top: total corpus articles per 0.25-wide bin of per-source mean rubric score, stacked by MBFC label group. Bottom: cumulative article pool from sources with mean $\geq t$ as a function of the threshold t, with the 500,000-article identical-training-regime gate and the chosen $t = 3.0$ marked ($t = 3.5$ falls below the gate).

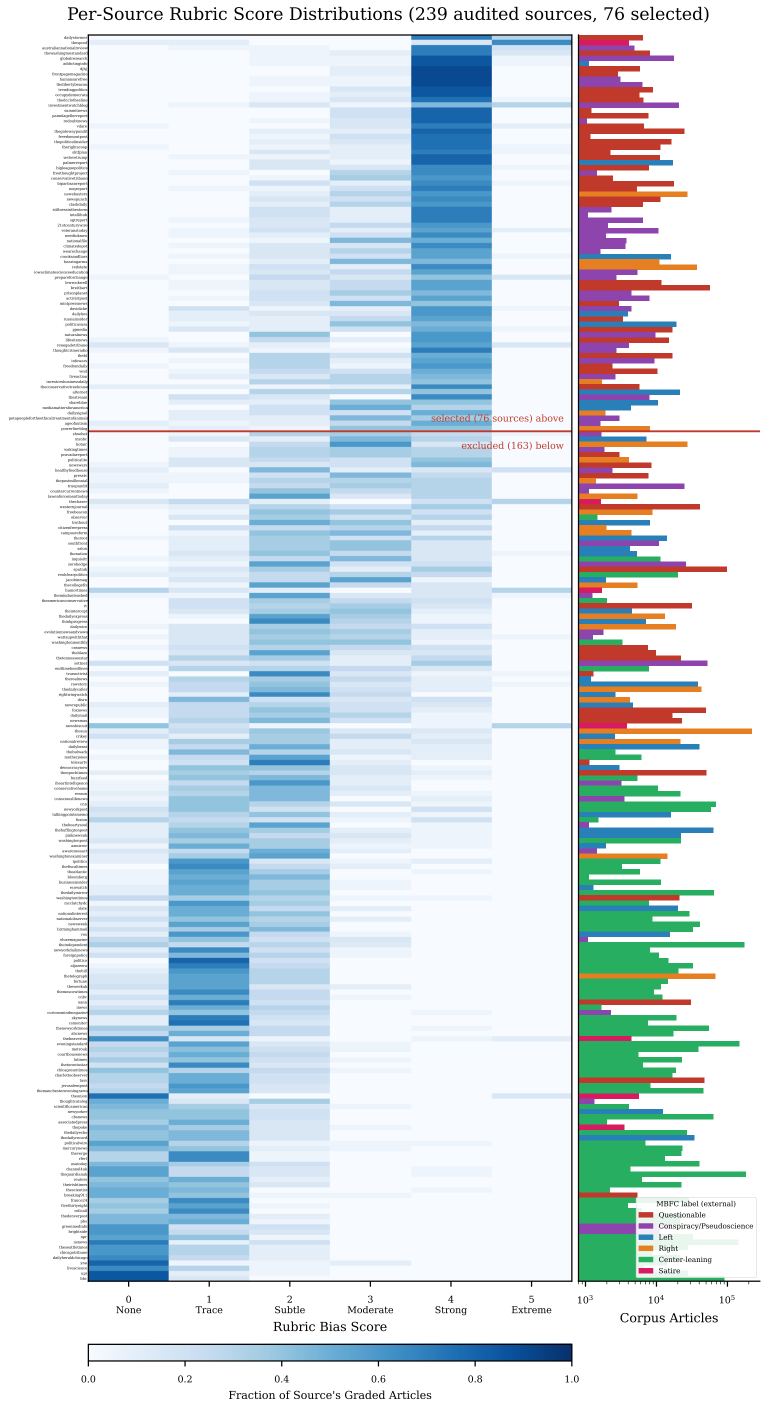

Figure 9: Per-source bias fingerprint. One row per audited NELA-GT source, sorted by per-source mean rubric score (descending). Left: fraction of each source's graded articles receiving each rubric score (0–5). Right: corpus article count per source (log scale), colored by MBFC label group. The red rule marks the mean ≥ 3.0 selection boundary.

H Code and Data Availability

All training, inference, scoring, and analysis code is available at `https://github.com/bxshan/research2026`, including: the SFT training script and the exact per-condition YAML configs (`model/`), the 15 inference prompts verbatim (`model/prompts.py`), the LLM-as-Judge pipeline and full rubric text (`data/bias_analysis/`), the GT-HB source-selection pipeline (`data/gt_hb/`: `select_sources.py`, `make_topup_list.py`, `finalize_sources.py`, and `sample_validation.py`), and the five WEAT word sets and permutation test (`weat/`). Training runs are reproduced via `model/run_cloud.sh` with the configs in `model/cfgs/`.

The NELA-GT and NELA-PS corpora are obtained from their maintainers [11, 4]; the Wikipedia high-school corpus can be rebuilt with the included download scripts (`data/full_download_scripts/`). The 1,500 scored completions with hallucination flags and the LoRA adapter weights for all four conditions are available from the authors on request.

I Licenses and Terms of Use for Existing Assets

This work builds entirely on existing public assets, used as follows. Wikipedia article text [5] is licensed under CC BY-SA 4.0; the derived high-school corpus is used for research only and is not redistributed. The NELA-GT articles are obtained through the misinfo-general distribution [11], released under CC BY-NC-SA 4.0 with gated access, as the original NELA-GT releases [3] have been deaccessioned from Harvard Dataverse; in line with its terms and the maintainers' explicit recommendation against openly sharing derivatives of the data, the adapters fine-tuned on this corpus are available upon reasonable request. Derived per-source statistics and judge scores used for GT-HB source selection, which contain no article text, are released in the code repository under the same CC BY-NC-SA 4.0 terms with attribution; the underlying articles are not redistributed. The NELA-PS corpus [4] is distributed via Harvard Dataverse (DOI `10.7910/DVN/YHWTFC`) under CC BY-NC 4.0. The base model, Llama 3.2 3B Instruct [6], is used under the Llama 3.2 Community License, which permits research use and fine-tuning. Bias scoring uses the Claude Haiku 4.5 API [12] under Anthropic's Commercial Terms of Service. Training and evaluation code builds on the HuggingFace `transformers` and `peft` libraries (Apache 2.0).

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